Frank Santaguida

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Skills

OpenAI APIs | RAG | MCP | LangChain | Pinecone | Python | TypeScript | Next.js | React | AWS | SQL

Experience

AI Software Engineer | Penn State University | PA | Feb 2023 – May 2026

- Architected and built **MaterialsTube**, an **NSF**-funded AI-powered video platform for the national materials science community, automating transcription, semantic search, and metadata generation.
- Delivered a full **RAG** pipeline for video data using **AWS**, transcript chunking, **OpenAI embeddings**, **Pinecone** vector search, **LangChain** orchestration, and streaming LLM responses.
- Engineered the LiST agentic chat interface using **OpenAI** Responses API, a **ReAct**-style agent loop, and **OpenAPI**-derived tool schemas, enabling structured tool-calling workflows for scientific datasets.
- Designed modular system prompts and safety guardrails to constrain agent behavior, reduce unintended mutations, and support safer interaction with production research data.
- Implemented a **FastMCP** server exposing **LiST** functionality through the **Model Context Protocol**, enabling external **MCP**-compatible clients to interact with platform tools.
- Rewrote a legacy **Knockout.js** research application into a modern **Next.js**, **React**, and **TypeScript SPA** while maintaining production continuity for active research users.
- Deployed and maintained **AWS** serverless infrastructure across **Lambda**, **DynamoDB**, **S3**, **CloudFormation**, **Cognito**, **AppSync GraphQL**, and **Amplify**, supporting AI transcription, vector processing, API integrations, and data pipelines.
- Designed **LLM-as-judge** evaluation harnesses to benchmark agent responses for correctness, speed, cost, and regression behavior against deterministic API ground truth.
- Developed a **Next.js** and **FastAPI** hybrid service for scientific data processing, including **AFM** image normalization, **XRD** peak detection, and **D3** visualizations for multi-modal materials characterization workflows.
- Applied **ADA** and **WCAG** standards across public-facing systems.
- Led internal AI education initiatives, translating emerging AI concepts into presentations, reasoning-flow diagrams, implementation guidance, and stakeholder-facing technical explanations.
- Served on the Materials Research Institute Staff Advisory Committee, participating in hiring, candidate evaluation, and institutional technology discussions.

Web Developer | 1SEO Digital Agency | PA | 2022 – 2023

- Developed and optimized client-facing web applications using **JavaScript**, **TypeScript**, **React**, **React Router**, **WordPress**, **Elementor**, **Divi**, **WooCommerce**, **Google Analytics** and technical SEO.
- Acted as technical lead for key accounts, translating client requirements into engineering scopes and custom front-end solutions.

Engineer / Consultant | Built by Frank | Philadelphia, PA | 2012 – 2022

- Led full project lifecycles for diverse clients, spanning discovery, architecture, database design, security implementation, full-stack development, deployment, and long-term support.
- Directed an international development team through agile workflows, coordinating implementation tasks, delivery timelines, technical reviews, and client reporting.
- Consulted on product roadmaps and delivered technical presentations to enterprise stakeholders and investors, aligning engineering decisions with business objectives.
- Developed a **Progressive Web App** to manage complex real estate continuing education credentials, user workflows, and administrative reporting.
- Implemented custom **Zoho CRM** integrations and data pipelines to map, validate, transform, and synchronize data from **RESTful APIs** and web-scraped sources.
- Built integrations across **Cloudflare**, **Google Maps API**, **WooCommerce**, **Google Analytics**, and third-party business platforms, including **webhooks**, authentication, and data synchronization.
- Deployed and managed cloud infrastructure using **Heroku**, **Netlify**, and automated **CI/CD** workflows while supporting delivery through documentation and client technical training.

Market Operations Consultant | Bravosolution, now Jaggaer | Malvern, PA | 2011

- Optimized sourcing and spend-management practices, capturing millions of dollars in value through improved data visibility across procurement events.
- Managed **RFQ**s, **B2B** auctions, supplier/client relationships, help-desk process changes, documentation, and auction events.

ERP & Accounting Specialist | Deacom | Wayne, PA | 2010

- Supported **DEACOM ERP** software across accounting modules, database management, production support, and general ledger discrepancy resolution.
- Resolved critical production-down emergencies and administered **Microsoft SQL Server** configurations, backups, restorations, and **SQL** scripts.

Associate ERP Support Analyst | Infor | Malvern, PA | 2008 – 2010

- Supported **Adage** and **MyDay ERP** software as a direct technical point of contact for major enterprise clients.
- Engineered custom **SQL** scripts to resolve emergency client database issues, correct production data problems, and support business-critical operations.
- Provisioned isolated staging environments using virtual machines, **Apache Tomcat**, **SQL** databases, and application configuration workflows.
- Owned alpha release testing for MyDay ERP software and authored internal technical documentation supporting enterprise support workflows.

Researcher & Developer | Pennsylvania State University | State College, PA | 2008
The Network-Centric Cognition and Information Fusion Center

- Designed a prototype UI for Penn State's security command center using **Adobe Flex** and **MVC**-based application design principles.
- Developed data-driven security solutions and technical proposals supporting campus safety, including a technical lexicon for NC2IF research workflows.

Education

BS in Information Sciences & Technology | Pennsylvania State University | 2008